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NAME :
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EXPERIMENT: FREE FALLING OBJECT

Objectives :

1. Determine the magnitude of the Earth's gravitational acceleration through the analysis of free-fall.

EXPERIMENT SUMMARY

INTRODUCTION

Motion is one of the most fundamental studies in physics that has been studied since Ancient Greece. In classical mechanics, the motion of an object is classified according to its acceleration characteristics. Free-fall motion is one of the most fundamental examples of motion with constant acceleration in classical mechanics. Before the 17th century, the generally accepted view was based on Aristotle's teachings that heavier objects fell faster than lighter objects. This view was eventually shattered by Galileo Galilei (1564–1642), who through a series of experiments on the oblique plane concluded that all objects fall at the same velocity at the same location, regardless of mass, as long as air resistance is ignored. Galileo's contribution became the foundation for the modern understanding of free-fall kinematics.

Free-fall motion is defined as the motion of an object that moves freely only under the influence of gravity, without air resistance, in the direction of vertical motion [1]. In these conditions, the object experiences a constant acceleration called Gravitational acceleration g and denoted by g . Giancoli [2] states that on the earth's surface, the magnitude is about 9.80 m/s^2 , although the value varies slightly depending on the altitude and geographical latitude of the measurement location. Halliday, Resnick & Walker [3] add that for the purposes of analysis, gravitational acceleration is treated as a downward-directed constant acceleration, this acceleration is expressed as $a = -g$. Since free-fall motion is motion with constant acceleration, the kinematic equation for constant acceleration is fully applicable. For an object that is dropped from a stationary state ($v_0 = 0$), the vertical position of the object as a function of time is expressed as [3]:

$$h = \frac{1}{2}gt^2 \quad (1)$$

This equation shows that height (h) is directly proportional to the square of time (t^2), so that plotting h against t^2 will produce a straight line passing through the origin with a gradient (m) = $\frac{g}{2}$. Therefore, the value of the gravitational acceleration can be calculated using the equation:

$$g = 2m \quad (2)$$

Abdullah [4] showed that the strength of the gravitational field on the earth's surface can theoretically be calculated from the universal gravitational constant (G), the mass of the earth (M_B), and the radius of the earth (R_B) through:

$$g = G \frac{M_B}{R_B^2} = 6,67 \times 10^{-11} \times \frac{5,98 \times 10^{24}}{(6,37 \times 10^6)^2} \approx 9,8 \text{ N/kg} \quad (3)$$

These results are consistent with widely accepted experimental values. It should be noted that Equation (1) does not contain the mass variable of the object, which explicitly confirms Galileo's finding that the acceleration of gravity is universal and does not depend on the mass of the falling object [2]. In this experiment, the validity of this statement was tested directly by varying the mass of the dropped ball, and the data obtained were also used to determine the experimental relationship between the falling height and the travel time.

In this experiment, the quality of measurement results is evaluated through calculations of relative error to the value of the theoretical gravitational acceleration $g_{\text{theoretical}} = 9,8 \text{ m/s}^2$. The relative error is calculated using the equation:

$$\text{Error} = \left| \frac{g_{\text{calculation}} - 9,8}{9,8} \right| \times 100\% \quad (4)$$

A small error value indicates that the experimental result are close to a theoretical value, while a large error indicates the presence of factors that affect the measurement during the experiment.

METHODOLOGY

A. Tools and Materials

The following are the tools and materials used in this experiment:

Table 1. Tools and materials for free-falling object.

Apparatus	Quantity	Function
Set Leybold free falling object	1 set	Used as a ball drop instrument, drop height gauge, and stopwatch until it touches the sensor.
Large iron ball	1 piece	Used as a dropped object.
Small iron ball	1 piece	Used as a dropped object.

B. Procedure

The following are the working steps in the free falling object experiment:

1. Arrange the equipment as Figure 1.

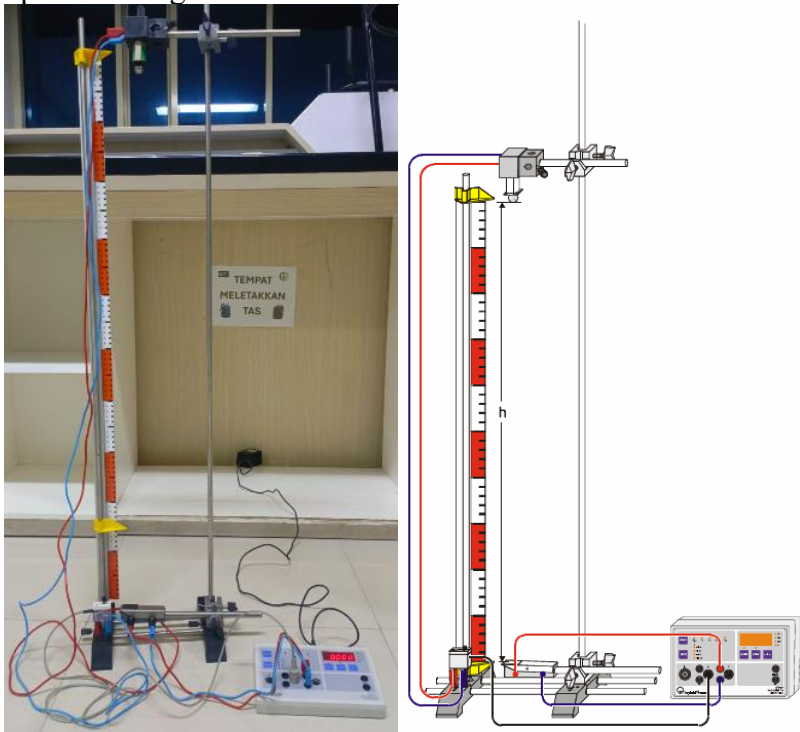


Figure 1. experiment tool scheme
Source: personal documents.

2. Connect the adapter to the power source.
3. Set the height (h) of the bottom of the stand. The ruler reading is ensured to start from 0 cm at the bottom of the stand as a reference for height measurement.
4. Attach the iron ball to the magnetic clamping as shown in Figure 2.

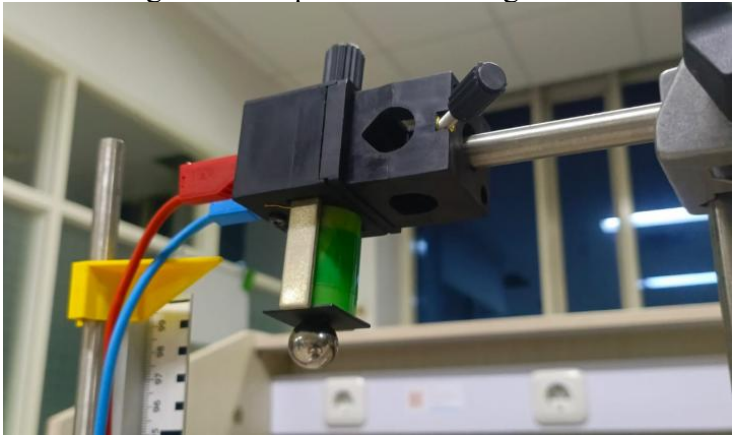


Figure 2. Magnetic clamping parts.
Source: personal documents.

5. Set the mode as shown in Figure 3, then press the *start button*.

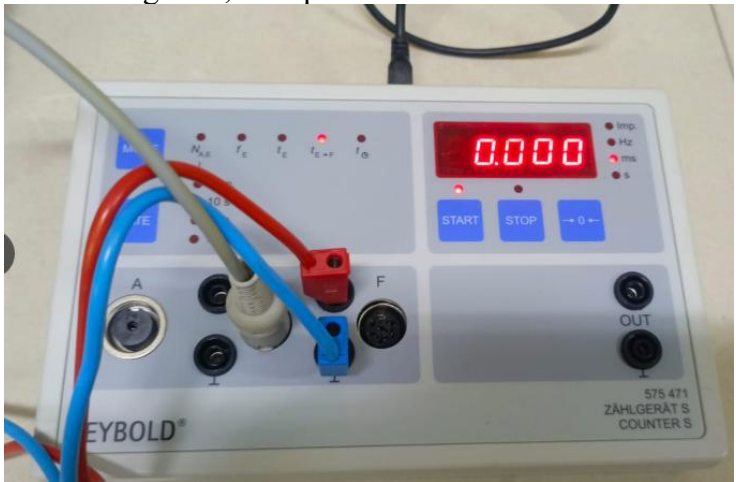


Figure 3. Leybold mode options
Source: personal documents.

6. Press the switch button in Figure 4(a) to drop the ball and automatically the timer starts counting and stops when the ball touches the contact plate as in Figure 4(b).

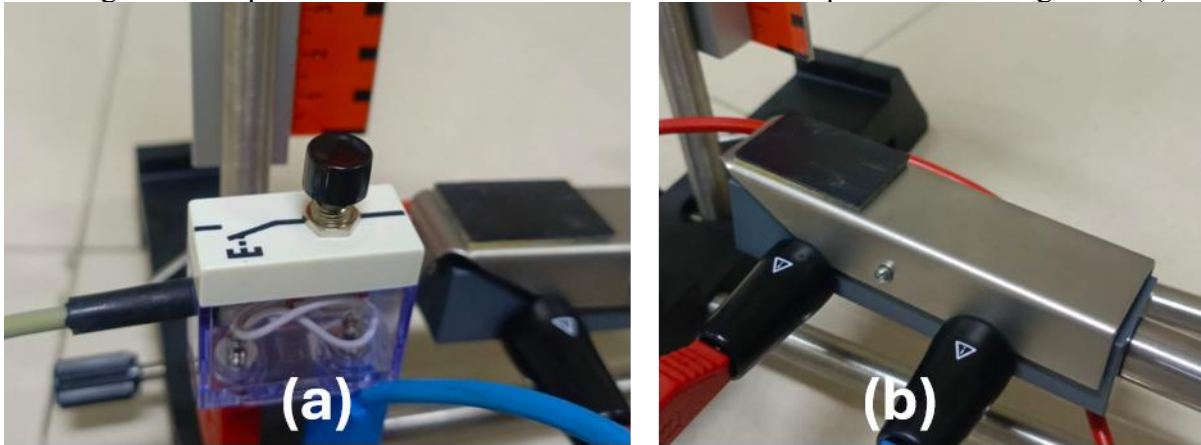


Figure 4. (a) Switch to drop the ball. (b) contact plate
Source: personal documents.

7. Record the time indicated by Leybold as shown in Figure 5 in the measurement results table.

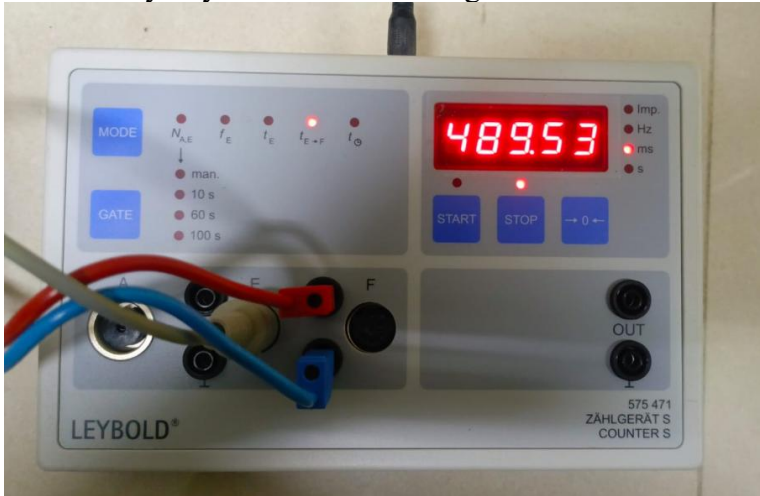


Figure 5. Example of time measurement results on Leybold
Source: personal documents.

8. Repeat steps 3 – 7 for heights of 80 cm, 90 cm and 100 cm, with 5 repetitions for each height variation.
9. Steps 3 – 8 are repeated using iron balls of different sizes.

E. Experiment Flowchart

DATA ANALYSIS AND DISCUSSION

A. Analysis of Measurement Data

The following is a table of data from the results of measurements made in the laboratory:

Table 2. The results of the measurement of the time of the fall of a small ball.

Repetition to-	t (ms)		
	0.8 meters	0.9 meters	1 meter
1			
2			
3			
4			
5			
Average			

Table 3. The measurement results of the time of the fall of a large ball.

Repetition to-	t (ms)		
	0.8 meters	0.9 meters	1 meter
1			
2			
3			
4			
5			
Average			

B. Calculation Analysis

Example calculation:

The following is a table of the data obtained from the calculations:

Table 4. The results of gravity calculations use small spheres.

Repetition to-	$g \text{ (m/s}^2\text{)}$		
	0.8 meters	0.9 meters	1 meter
1			
2			
3			
4			
5			
Average			

Table 5. The results of gravity calculations use large spheres.

Repetition to-	$g \text{ (m/s}^2\text{)}$		
	0.8 meters	0.9 meters	1 meter
1			
2			
3			
4			
5			
Average			

Table 6. The result of the calculation of the gravitational error of a small sphere.

Repetition to-	Percentage <i>error</i> (%)		
	0.8 meters	0.9 meters	1 meter
1			
2			
3			
4			
5			
Average			

Table 7. The result of the calculation of the gravitational error of a large sphere.

Repetition to-	Percentage <i>error</i> (%)		
	0.8 meters	0.9 meters	1 meter
1			
2			
3			
4			
5			
Average			

C. Graphics

(Graph the height () of the rope against the square period (th^2) and calculate the acceleration of gravity based on the fork)

D. Discussion

CONCLUSION

From this experiment it can be concluded:

- 1.

REFERENCE

- [1] R. A. Serway and J. W. Jewett, *Physics for Scientists and Engineers with Modern Physics*, 9th ed. Mason, OH: Brooks/Cole, Cengage Learning, 2012.
- [2] D. C. Giancoli, *Physics: Principles with Applications*, 7th ed. Upper Saddle River, NJ: Pearson Education, 2014.
- [3] D. Halliday, R. Resnick, and J. Walker, *Fundamentals of Physics Extended*, 9th ed. Hoboken, NJ: John Wiley & Sons, 2010.
- [4] M. Abdullah, *Basic Physics I*, Revised Edition. Bandung: ITB Publishers, 2016.

PRELIMINARY ASSIGNMENT

1. Mention and explain two examples of the application of the concept of free fall in daily life.
2. A ball is dropped from a height with no initial velocity. Derive the equation used to determine the acceleration of gravity based on the relationship between the height of the fall and the travel time, and then explain why the graph against h was chosen as an analysis method over the graph against t .